Task 1

A satellite structure with a parabolic antenna mounted on a truss (see sketch) shall be analyzed with respect to its dynamic behavior. The system is composed of three struts, each with a stiffness k. The mass is lumped into one single mass m in the connecting node. Two translational degrees of freedom shall be examined. There is an external test load F acting.

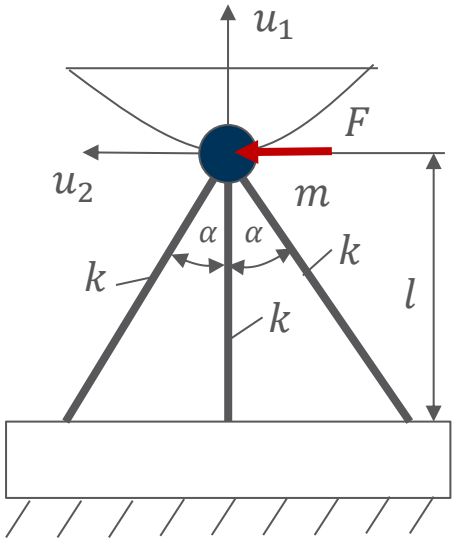
a) Determine the vibrational equation of motion using the Lagrange formalism. Determine the mass and stiffness matrix of the system.
b) Determine all eigenfrequencies of the system.
c) Determine and sketch all eigenmodes of the system.
d) Determine the mass normalized modal matrix and formulate the normalized differential equation with modal coordinates.

$$k = 1$$

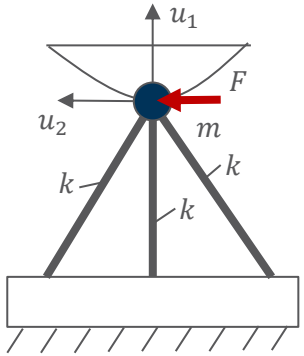
$$m = \frac{3}{2}$$

$$\alpha = 30°$$

$$l = 1$$



Task 1



a) Determine the vibrational equation of motion using the Lagrange formalism. Determine the mass and stiffness matrix of the system.

$L = K - P \qquad q_1 = u_1 \quad q_2 = u_2$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \left(\frac{\partial L}{\partial q_j}\right) = \frac{d}{dt}\left(\frac{\partial K}{\partial \dot{q}_j} - \frac{\partial P}{\partial \dot{q}_j}\right) - \left(\frac{\partial K}{\partial q_j} - \frac{\partial P}{\partial q_j}\right) = Q = \frac{\partial(\delta W)}{\partial(\delta q_j)}$$

$$K = \frac{1}{2} m \dot{u}_1{}^2 + \frac{1}{2} m \dot{u}_2{}^2$$

$$P = \frac{1}{2} k u_1^2 + \frac{1}{2} k (u_1 \cos(30°))^2 + \frac{1}{2} k (u_1 \cos(30°))^2 + \frac{1}{2} k (u_2 \sin(30°))^2 + \frac{1}{2} k (u_2 \sin(-30°))^2 =$$

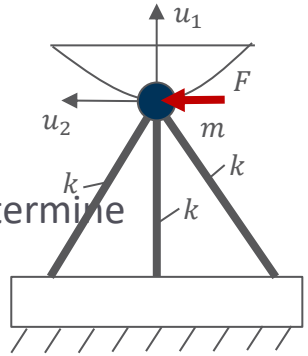
$$P = \frac{1}{2} k u_1^2 + \frac{1}{2}\frac{3}{4} k u_1^2 + \frac{1}{2}\frac{3}{4} k u_1^2 + \frac{1}{2}\frac{1}{4} k u_2^2 + \frac{1}{2}\frac{1}{4} k u_2^2 =$$

Refer to last slide for more details.

$$P = \frac{1}{2} k u_1^2 + \frac{3}{4} k u_1^2 + \frac{1}{4} k u_2^2 = \frac{5}{4} k u_1^2 + \frac{1}{4} k u_2^2$$

$$\delta W = F \delta u_2$$

Task 1



a) Determine the vibrational equation of motion using the Lagrange formalism. Determine the mass and stiffness matrix of the system.

$$\frac{\partial K}{\partial q_j} = 0 \qquad \frac{\partial P}{\partial \dot{q}_j} = 0$$

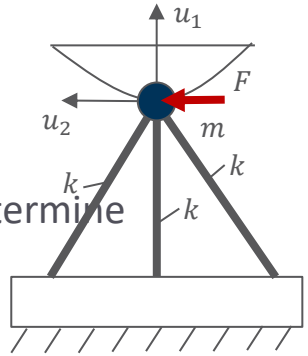
$$\frac{\partial P}{\partial u_1} = \frac{5}{2} k u_1 \qquad \frac{\partial P}{\partial u_2} = \frac{1}{2} k u_2$$

$$\frac{d}{dt}\left(\frac{\partial K}{\partial \dot{u}_1}\right) = m\ddot{u}_1 \qquad \frac{d}{dt}\left(\frac{\partial K}{\partial \dot{u}_2}\right) = m\ddot{u}_2$$

$$\frac{\partial(\delta W)}{\partial(\delta u_1)} = 0 \qquad \frac{\partial(\delta W)}{\partial(\delta u_2)} = F$$

$$m\ddot{u}_1 + \frac{5}{2} k u_1 = 0 \qquad m\ddot{u}_2 + \frac{1}{2} k u_2 = F$$

Task 1



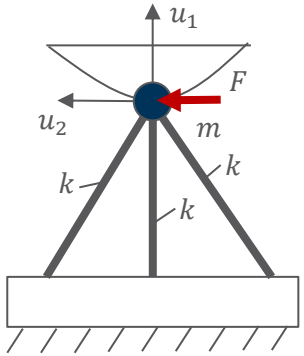
a) Determine the vibrational equation of motion using the Lagrange formalism. Determine the mass and stiffness matrix of the system.

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{Bmatrix} + \begin{bmatrix} \frac{5}{2}k & 0 \\ 0 & \frac{1}{2}k \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ F \end{Bmatrix}$$

$$\frac{3}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} \ddot{u}_1 \\ \ddot{u}_2 \end{Bmatrix} + \frac{1}{2}\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ F \end{Bmatrix}$$

$$[M] = \frac{3}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad [K] = \frac{1}{2}\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$$

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b) Determine all eigenfrequencies of the system

$$[M]\{\ddot{u}\} + [K]\{u\} = \{0\} \qquad \{u\} = \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix}$$

$$\frac{3}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\{\ddot{u}\} + \frac{1}{2}\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}\{u\} = \{0\}$$

$$\det([K - \lambda M]) = \{0\}$$

$$\det\left(\begin{bmatrix} \frac{5}{2} - \frac{3}{2}\lambda & 0 \\ 0 & \frac{1}{2} - \frac{3}{2}\lambda \end{bmatrix}\right) = \{0\}$$

$$\lambda_1 = \frac{1}{3} \qquad \omega_1 = \sqrt{\lambda_1} \qquad f_1 = \frac{\omega_1}{2\pi} = 0.090Hz$$

$$\lambda_2 = \frac{5}{3} \qquad \omega_2 = \sqrt{\lambda_2} \qquad f_2 = \frac{\omega_1}{2\pi} = 0.205\ Hz$$

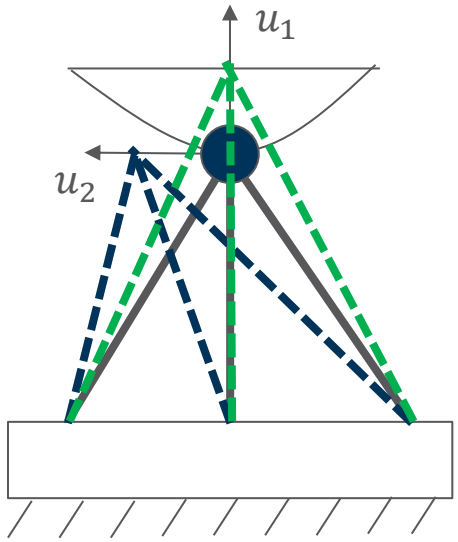
Result of a)

$$[M] = \frac{3}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$[K] = \frac{1}{2}\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$$

Task 1

c) Determine and sketch all eigenmodes of the system.



$$([K - \lambda_i M])\{\Phi_i\} = \{0\}$$

$$\begin{bmatrix} \frac{5}{2} - \frac{3}{2}\frac{1}{3} & 0 \\ 0 & \frac{1}{2} - \frac{3}{2}\frac{1}{3} \end{bmatrix} \begin{Bmatrix} \Phi_{1,1} \\ \Phi_{1,2} \end{Bmatrix} = 0$$

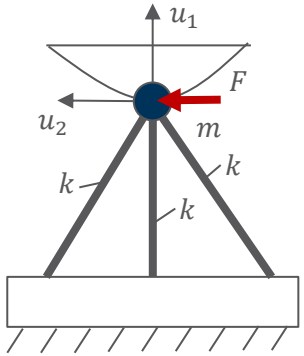
$$\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} \Phi_{1,1} \\ \Phi_{1,2} \end{Bmatrix} = 0$$

$$\{\Phi_1\} = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$$

$$\{\Phi_2\} = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$$

$$\Phi = [\Phi_1 \quad \Phi_2] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Task 1



d) Determine the mass normalized modal matrix and formulate the normalized differential equation with modal coordinates.

$$[\Phi]^T[M][\Phi] = \begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\frac{3}{2}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}\gamma_1 & 0\\ 0 & \gamma_2\end{bmatrix}$$

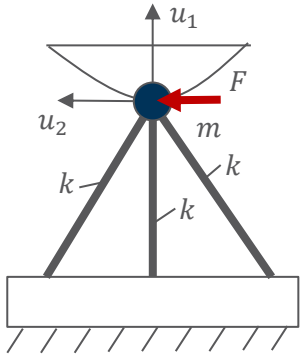
$$\begin{bmatrix}\gamma_1 & 0\\ 0 & \gamma_2\end{bmatrix} = \begin{bmatrix}\frac{3}{2} & 0\\ 0 & \frac{3}{2}\end{bmatrix}$$

$$\Phi_{i,n} = \frac{1}{\sqrt{\gamma_i}}\Phi_i$$

$$\Phi_{1,n} = \sqrt{\frac{2}{3}}\begin{Bmatrix}0\\ 1\end{Bmatrix} \qquad \Phi_{2,n} = \sqrt{\frac{2}{3}}\begin{Bmatrix}1\\ 0\end{Bmatrix}$$

$$[\Phi_n] = \sqrt{\frac{2}{3}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}$$

Task 1



d) Determine the mass normalized modal matrix and formulate the normalized differential equation with modal coordinates.

$$\frac{3}{2}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\{\ddot{x}\}+\frac{1}{2}\begin{bmatrix}5 & 0\\ 0 & 1\end{bmatrix}\{x\}=\{F\}$$

$$\{x\}=[\Phi]\{q\}$$

$$[\Phi_n]^T\,[M][\Phi_n]\{\ddot{q}\}+[\Phi_n]^T[K][\Phi_n]\{q\}=[\Phi_n]^T\{F\}$$

$$\sqrt{\frac{2}{3}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\frac{3}{2}\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\sqrt{\frac{2}{3}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\{\ddot{q}\}+\sqrt{\frac{2}{3}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\frac{1}{2}\begin{bmatrix}5 & 0\\ 0 & 1\end{bmatrix}\sqrt{\frac{2}{3}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\{q\}=\sqrt{\frac{2}{3}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{Bmatrix}0\\ F\end{Bmatrix}$$

$$\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\{\ddot{q}\}+\begin{bmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{bmatrix}\{q\}=\sqrt{\frac{2}{3}}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix}\begin{Bmatrix}0\\ F\end{Bmatrix}$$

Appendix

